Undergraduate Texts in Mathematics

Editors
J.H. Ewing
F.W. Gehring
P.R. Halmos

Undergraduate Texts in Mathematics

Apostol: Introduction to Analytic Number Theory. Second edition.
Armstrong: Groups and Symmetry.
Armstrong: Basic Topology.
Bak/Newman: Complex Analysis.
Banchoff/Wermer: Linear Algebra Through Geometry. Second edition.
Berberian: A First Course in Real Analysis.
Brémaud: An Introduction to Probabilistic Modeling.
Bressoud: Factorization and Primality Testing.
Bressoud: Second Year Calculus.
Readings in Mathematics.
Brickman: Mathematical Introduction to Linear Programming and Game Theory.
Cederberg: A Course in Modern Geometries.
Childs: A Concrete Introduction to Higher Algebra.
Chung: Elementary Probability Theory with Stochastic Processes. Third edition.
Cox/Little/O'Shea: Ideals, Varieties, and Algorithms.
Curtis: Linear Algebra: An Introductory Approach. Fourth edition.
Devlin: The Joy of Sets: Fundamentals of Contemporary Set Theory. Second edition.
Dixmier: General Topology.
Driver: Why Math?
Ebbinghaus/Flum/Thomas: Mathematical Logic.
Edgar: Measure, Topology, and Fractal Geometry.
Fischer: Intermediate Real Analysis.
Flanigan/Kazdan: Calculus Two: Linear and Nonlinear Functions. Second edition.
Fleming: Functions of Several Variables. Second edition.
Foulds: Optimization Techniques: An Introduction.
Foulds: Combinatorial Optimization for Undergraduates.
Franklin: Methods of Mathematical Economics.
Halmos: Finite-Dimensional Vector Spaces. Second edition.
Halmos: Naive Set Theory.
Hämmerlin/Hoffmann: Numerical Mathematics.
Readings in Mathematics.
Iooss/Joseph: Elementary Stability and Bifurcation Theory. Second edition.
James: Topological and Uniform Spaces.
Jänich: Topology.
Klambauer: Aspects of Calculus.
Kinsey: Topology of Surfaces.
Lang: A First Course in Calculus. Fifth edition.
Lang: Calculus of Several Variables. Third edition.
Lang: Introduction to Linear Algebra. Second edition.
Lang: Linear Algebra. Third edition.
Lang: Undergraduate Algebra. Second edition.
Lang: Undergraduate Analysis.
Lax/Burstein/Lax: Calculus with Applications and Computing. Volume 1.
LeCuyer: College Mathematics with APL.

(continued after index)

George Pedrick

A First Course in Analysis

With 62 Illustrations



Springer Science+Business Media, LLC

George Pedrick
6231 Fairlane Drive
Oakland, CA 94611
USA

Editorial Board

J.H. Ewing
Department of
Mathematics
Indiana University
Bloomington, IN 47401
USA

F.W. Gehring
Department of
Mathematics
University of Michigan
Ann Arbor, MI 48019
USA

P.R. Halmos
Department of
Mathematics
Santa Clara University
Santa Clara, CA 95053
USA

Mathematics Subject Classifications (1991): 26-01

Library of Congress Cataloging-in-Publication Data
Pedrick, George.
A first course in analysis / George Pedrick.
p. cm. — (Undergraduate texts in mathematics)
Includes bibliographical references.
ISBN 978-1-4612-6435-4 ISBN 978-1-4419-8554-5 (eBook)
DOI 10.1007/978-1-4419-8554-5
1. Mathematical analysis. I. Title. II. Series.
QA300.P38 1994
515—dc20 93-5141

Printed on acid-free paper.

© 1994 Springer Science+Business Media New York
Originally published by Springer-Verlag New York in 1994
Softcover reprint of the hardcover 1st edition 1994

All rights reserved. This work may not be translated or copied in whole or in part without the written permission of the publisher (Springer Science+Business Media, LLC), except for brief excerpts in connection with reviews or scholarly analysis. Use in connection with any form of information storage and retrieval, electronic adaptation, computer software, or by similar or dissimilar methodology now known or hereafter developed is forbidden.
The use of general descriptive names, trade names, trademarks, etc., in this publication, even if the former are not especially identified, is not to be taken as a sign that such names, as understood by the Trade Marks and Merchandise Marks Act, may accordingly be used freely by anyone.

Production coordinated by Brian Howe and managed by Bill Imbornoni; manufacturing supervised by Vincent Scelta.
Typeset by Asco Trade Typesetting Ltd., Hong Kong.

9 8 7 6 5 4 3 2 1

To
Katie and Alexandra

Preface

The first course in Analysis, which follows calculus, along with other courses, such as differential equations and elementary linear algebra, in the curriculum, presents special pedagogical challenges. There is a change of stress from computational manipulation to "proof." Indeed, the course can become more a course in Logic than one in Analysis. Many students, caught short by a weak command of the means of mathematical discourse and unsure of what is expected of them, what "the game" is, suffer bouts of a kind of mental paralysis.

This text attempts to address these problems in several ways:

First, we have attempted to define "the game" as that of "inquiry," by using a form of exposition that *begins with a question* and proceeds to analyze, ultimately to answer it, bringing in definitions, arguments, conjectures, examples, etc., as they arise naturally in the course of a narrative discussion of the question. (The true, historical narrative is too convoluted to serve for first explanations, so no attempt at historical accuracy has been made; our narratives are completely contrived.)

Second, we have kept the logic informal, especially in the course of preliminary speculative discussions, where common sense and plausibility—tempered by mild skepticism—serve to energize the inquiry. The role of formal logic is meant to emerge gradually, as the means to fulfill the *felt need for certainty*, which arises in the course of such tentative, plausible arguments. Having a view of the broad outlines and recognizing the gaps in the clarity and logical support, the stage of careful formulation of definitions and theorems, perhaps after testing and revision, leads to the substance of a concise "Definition–Theorem–Proof" exposition of the result. The reader should be able to extract such an exposition from ours, primarily by deleting most of the expository discussion.

Thus, we regard the habit of clarity and conciseness as a *goal*, to be developed gradually. A student's impatience with the style of the text could be taken as a measure of progress.

Third, we aim toward increasing facility in *all* aspects of mathematical discourse, not just step-by-step formal reasoning. (One who has followed every step in a proof does not, as a result, understand it necessarily; he is only convinced of its validity.) We include things like: fancier calculation; selection and use of appropriate notation and terminology; conjecture; awareness of applicability of previously proved results; counterexample/proof; suitable generalization; critical reading and explaining; as aspects of the maturing process which the text seeks to promote (along with the accumulation of information, of course).

It makes sense to speak of the "march of the reasoning" at two levels: the overall development of the subject, and the internal reasoning from step to step, from hypothesis to conclusion, in a proof. We have embraced the first of these more explicitly than other texts in the conviction that it is necessary to the understanding of the material and the maturing of the mind. We are also convinced that Analysis is the proper subject for the siting of the material in such a broader landscape.

A further pedagogically challenging aspect of this course is the collection of suitable exercises. Routine, scales and arpeggios type, exercises for the mastery of computational techniques are rarely available or of great interest; it is, rather, the skills of mathematical discovery and discourse that need to be practiced.

Within the confines of any single topic, only a small number of results exist that are in the first line of interest and some of those are too difficult to prove, so they are left as exercises. We have tried to preserve a good number of these for the purposes of practice. Of course, that means we have omitted the proofs at many places in our exposition. That would be a fault in a careful logical exposition, which must place heavy weight on completeness. We claim it as a virtue of our text because many good exercises result and the need to contrive exercises by calling for proofs of more obscure facts is lessened.

If the reader is able to fill a good number of the gaps we leave (whether or not we ask him to in an exercise) and if he is persuaded that he could fill out most of the rest with a good effort, the loss of logical completeness is a small price to pay.

The text can be used in several ways, both by a teacher and by an individual reader. The portion entitled Background—a narrative development of the number systems beginning with counting and arriving at the rationals—is meant to be read for notations and terminology and perhaps revisited as the work progresses. (There is a danger of bogging down in this material and not getting very far with the Analysis, the subject of the book.)

The course proper can begin either with the axioms of ordered fields (p. 19) or with Part I.

The standard material for a minimal background in Analysis: complete-

ness, compactness, continuity, etc., is treated in Part I. An exceptional feature is that different questions are pursued which lead to several formulations of the completeness property, which are then shown to be equivalent. Completeness, rather than having the appearance of an additional axiom from "somewhere," is seen to solve several interesting problems.

The inclusion of some initial sections from Part II along with most of Part I will achieve a good one-course introduction to Real Analysis, either as a terminal course or as background to more advanced Analysis (Lebesgue integration, complex variables, differential equations, manifolds, etc.).

The complete text is enough for a two-course sequence and, it is hoped, will be found to be enjoyable "further reading," whether in a formal course or not. A good treatment of the Stieltjes integral can easily be found, in, e.g., Ross [23], Widder [27], or Hille [14]. For that reason, we have left it out here, even though it would fit admirably in Part II.

Portions of the text evolved as classroom notes over several years of teaching the shorter course. In this shorter version my colleague Russell Merris taught from the notes and made helpful suggestions. Marion Quaas typed them and Ann Cambra typed the final entire text. I owe them all my sincere thanks.

Finally, I am deeply indebted to Don Albers for his help and encouragement.

Hayward, CA, 1993 GEORGE PEDRICK

Contents

Preface vii
Notations and Conventions xv
Introduction xix

BACKGROUND
Number Systems 1

§1. Counting: The Natural Numbers 1
§2. Measurement: The Rational Numbers 13
The Axioms of Ordered Fields 19
§3. Decimal Representation. Irrationals 25

PART I
ANALYSIS 35

CHAPTER 1
Approximation: The Real Numbers 37

§1. Least Upper Bound 38
§2. Completeness. Nested Intervals 40
§3. Bounded Monotonic Sequences 42
§4. Cauchy Sequences 47
§5. The Real Number System 50
§6. Countability 52
Appendix. The Fundamental Theorem of Algebra. Complex Numbers 55

CHAPTER 2
The Extreme-Value Problem 64

§1. Continuity, Compactness, and the Extreme-Value Theorem 65
§2. Continuity of Rational Functions. Limits of Sequences 72
Appendix. Completion of the Proof of the Fundamental Theorem of Algebra 76
§3. Sequences and Series of Reals. The Number e 78
§4. Sets of Reals. Limits of Functions 90

CHAPTER 3
Continuous Functions 97

§1. Implicit Functions. $\sqrt[n]{x}$. The Intermediate-Value Theorem 97
§2. Inverse Functions. x^r for $r \in \mathbb{Q}$ 101
§3. Continuous Extension. Uniform Continuity. The Exponential and Logarithm 104
§4. The Elementary Functions 109
§5. Uniformity. The Heine–Borel Theorem 112
§6. Uniform Convergence. A Nowhere Differentiable Continuous Function 118
§7. The Weierstrass Approximation Theorem 124
Summary of the Main Properties of Continuous Functions 128
Appendix. A Space-Filling Continuous Curve 128

PART II
FOUNDATIONS OF CALCULUS 131

CHAPTER 4
Differentiation 133

§1. Differential and Derivative. Tangent Line 135
§2. The Foundations of Differentiation 140
§3. Curve Sketching. The Mean-Value Theorem 146
§4. Taylor's Theorem 154
§5. Functions Defined Implicitly 160

CHAPTER 5
Integration 169

§1. Definitions. Darboux Theorem 171
§2. Foundations of Integral Calculus. The Fundamental Theorem of Calculus 179
§3. The Nature of Integrability. Lebesgue's Theorem 189
§4. Improper Integral 197
§5. Arclength. Bounded Variation 206
A Word About the Stieltjes Integral and Measure Theory 215

CHAPTER 6
Infinite Series 218

§1. The Vibrating String 219
§2. Convergence: General Considerations 224
§3. Convergence: Series of Positive Terms 230
§4. Computation with Series 237
§5. Power Series 244
§6. Fourier Series 254

Bibliography 266

Index 269

Notations and Conventions

The Text

Comparative terms are understood in the wide sense; "greater," for example, is understood to include the case of equality. In cases where the equality is to be excluded, the terminology used makes it explicit, e.g., "strictly greater" ($>$ or $\gneqq$ replaces $\geq$).

Definitions, which are sometimes presented formally, as they would be in a "Definition–Theorem–Proof" exposition, are also frequently embedded in more informal exposition (*even in exercises*). In such cases, the term is printed in **boldface** in the sentence which serves to define it precisely.

Quotation marks are used frequently (as in the above "Definition–Theorem–Proof" usage) to invoke concepts that are ill-defined or tentative, in the course of an informal exposition. In particular, discussions of the progress toward a precise, formal definition of a term might use this device.

This typeface is used to identify some parts of the exposition as being strictly motivational, parts that would surely be deleted if one were to seek to extract from the text a closer approximation of a "Definition–Theorem–Proof" presentation.

The appelation **Theorem** is reserved for those results that are judged to be part of the *goals* of the theory, as contrasted to **Lemmas** (technical results—used to consolidate proofs) and **Propositions** (everything else, including less central results, perhaps just pointing up relations between terms, e.g., open set = complement of closed set).

The Exercises

Some exercises are utterly trivial. The exercise is to recognize the fact. More generally, to assess the difficulty of each exercise is the first step in addressing it. An exercise that simply states something is understood to call for a proof of that statement.

An *overall exercise* is to rewrite portions of the text, with a view to achieving more "elegance" (brevity with clarity!), or a more clearly unassailable logic in the "march of the reasoning" as in a "Definition–Theorem–Proof" presentation.

A few exercises call for the formulation of precise definitions, or for the introduction of suitable notations, as reminders of the importance of these steps in the achievement of satisfactory theories.

Notations

The notations of set theory are by now quite standard and no lengthy explanation is deemed necessary. We only mention them briefly:

$\in$:	"is a member of" or "belongs to."
$\notin$:	is not a member of (use of / to negate).
$\subset$	"included in" or "is a subset of" (wide sense!).
$\{x \in S: ----\}$	"the set of all x belonging to S such that ----"
$\cup$:	union.
$\cap$:	intersection.
$-$:	complement: $X - S = \{x \in X: x \notin S\}$.
ϕ:	empty set.
disjoint:	having no members in common.
$\times$:	Cartesian product: $X \times Y = \{$ordered pairs (x, y): $x \in X$ and $y \in Y\}$.
$f: X \to Y$:	f is a function of $x \in X$ with values $y \in Y$.
$y = f(x)$ or $x \to f(x)$:	abbreviation of above.
dom f:	domain of the function $f = \{x \in X: f(x)$ defined$\}$.
ran f:	range of the function $f = \{y \in Y: y = f(x)$ for some $x\}$.
graph f:	graph of the function f:

$$\text{graph } f = \{(x, y) \in X \times Y: y = f(x)\}.$$

$F\|_S$:	restriction of the function f to S:

$$F|_S = \{(x, y) \in X \times Y: x \in S \cap \text{dom } f \text{ and } y = f(x)\}.$$

$f \circ g$:	composite of functions: g followed by f

$$z = [f \circ g](x) \qquad \text{if} \quad z = f(g(x)).$$

References

To formulas and figures, by numbering them within a discussion, are regarded as distractions and are kept to a minimimum, by either repeating the entire formula or leading the reader to hold the proper figure or formula in mind.

To other parts of the text are by chapter and section number, if not page.

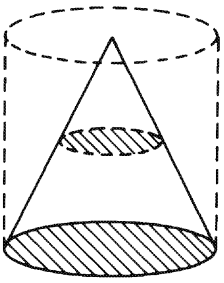
To the Bibliography are by author's name and item number, e.g., Lebesgue [20] refers to item 20 in the Bibliography, a book by H. Lebesgue.

Introduction

Ideas of infinity—the "infinitely large" and the "infinitely small" (or "infinitesimal")—are traceable in the history of human thought to the earliest times. It is both natural and useful to imagine procedures of a kind that can be repeated over and over (bisecting a line segment, for example, or extending one to twice its length) and this evokes at once the notion of repetition "infinitely often." In turn, notions of "infinite extent" and "infinitely divisible" arise.

The usefulness of such notions is tempered by the paradoxes to which their uncritical application can lead. Examples were recorded as early as the fifth century B.C. in Zeno's famous paradoxes. One of these argues that motion is impossible, via the infinite repetition of the reasoning, that one can only move from A to B by first moving from A to the half-way point, which requires moving from A to the quarter-way point, etc.; the infinite repetition postpones indefinitely the possibility of leaving A.

To cite another example, it is a compelling intuition to think of a cone as



a stack of infinitely many infinitely thin disks. Ancient thinkers used this intuition to discover (i.e., guess, then prove) that the volume of the cone is one-third that of the circumscribed cylinder—a substantial achievement. But surely two "adjacent" disks have the same radius, which implies—via the same compelling intuition—that a cone cannot exist!

Surrounded by mystery and paradox, the uses of infinity expanded through the centuries, a historical process which reached its culmination in the late-seventeenth century with the development of The Calculus, and in the eighteenth century with spectacular applications of the new techniques. The lingering mysteries loomed larger as more delicate matters were investigated. It came to be recognized that the resolution of these issues depended on a better clarification of the two most fundamental concepts in mathematics: number and function.

By the end of the nineteenth century, all of the difficulties had been cleared up and it could be said that "infinity" was well understood. This set the stage for the extraordinary explosion of knowledge witnessed by the twentieth century.

It is this nineteenth-century work which this book undertakes to discuss. It was a crucial transitional body of material in history and it is a crucial transitional subject in the modern mathematics curriculum. The student will find that the focus of attention is shifted from the learning and applying of computational techniques to careful reasoning from hypothesis to conclusion, in its various forms. (It should be pointed out, however, that even the most routine computation is just a highly abbreviated presentation of an argument, or "proof," e.g., the "solution"

$$x^2 - 2x - 3 = 0,$$

$$(x + 1)(x - 3) = 0,$$

$$x + 1 = 0 \quad \text{or} \quad x - 3 = 0,$$

$$x = -1 \quad \text{or} \quad x = 3,$$

is just a sketch of an argument which proves that the only numbers x for which $x^2 - 2x - 3$ is zero are the numbers -1 and 3. The transition, therefore, is one of stress; the arguments, being less routine, demand more attention individually.)

In the old days, a function was a formula or a temperature or a velocity. The concept of function was loaded with all of these extraneous connotations. The clarification of this concept was simply the dropping of the extraneous ideas until there remained the abstract idea of a many-to-one correspondence. Modern students are taught the function concept in its full abstract form from their early years, so what was once a stumbling block is no longer.

The number concept is more complicated. Each person has an extensive familiarity with numbers and their uses, perhaps some uncertainties as well.

One of the achievements of the nineteenth century was the recognition that the proper objects of attention were not "numbers," but "number systems." From the modern perspective, we can imagine a process of evolution of "number systems," each stage being a natural response to clearly perceived defects in the previous stage. We provide a sketch of such an evolution as background for our text, at least to the point at which the perceived defects—various forms of "incompleteness" of the number systems—become central to our subject, and their correction the first major topic of that subject: Analysis.

Of course, these number systems contain no numbers that are "infinite" or "infinitesimal"; it is not in that direction that the new understanding was to be found. It is, instead, *sets* of numbers that provide the key, together with the use of *quantified* statements about the numbers in the sets. There are two kinds of quantified statements about a member x of a set of things S:

The universal quantifier—expressed as "For every x in S ..." or "For any x in S ..." or "For each x in S ..." (the ... being filled by an assertion about x).

The existential quantifier—expressed as "There exists an x in S such that ..." or "For some x in S ...".

It turns out that every useful statement about the infinite or infinitesimal has a precise formulation as a quantified statement about a number in a set that contains, respectively, arbitrarily large or arbitrarily small numbers. Much of the content of this book consists of a setting forth of the details of how this "turns out" to be true.

Finally, it can be claimed that the arguments that achieved these peaks of nineteenth-century clarification are well on their way to becoming the standard devices for new developments in mathematics and its applications ... to becoming clichés in engineering, numerical analysis, and similar areas, just as the quadratic equation or integration-by-parts once did.